Technology Providers of Synthetic Inertia

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Abstract

The Australian grid is currently transitioning away from carbon based energy generation towards renewable sources, such as wind and solar. As a result, the power system has and continues to see increasing integration of inverter-based resources (IBR), significantly impacting the dynamics and stability of power systems. One concern is the increasing rate of change of frequency (RoCoF) as a consequence of the reduction in system synchronous inertia, an inherent trait of synchronous generator (SG) electromechanical coupling to the grid. This paper acknowledges the importance of Grid-Forming (GFM) converters supported by lithium-ion based battery energy storage systems (BESS), as a technology to support grid stability through a synthetic inertial response (SIR). Upon a literature review, this report evaluates key parameters affecting the capability of BESS to provide SIR; Inverter control algorithms,virtual inertia (H) and damping (D) constants, BESS State of Charge (SoC), BESS operating point (OP), inverter overload capability, and grid SCR and X/R ratio. EMT modelling of a single BESS+GFM converter infinite bus model was performed with MATLAB Simulink to identify parametric relationships between the aforementioned characteristics and consequential response of the system during load, frequency and angle changes.

1 Introduction

There is currently a global increase in the integration of variable renewable energy (VRE) sources in modern power systems. These inverter-based resources (IBRs), led by solar and wind generation, are progressively replacing traditional synchronous generators (SGs) as the primary source of power generation.The evolution in energy generation technology is a response to global climate and sustainability goals to counter excessive pollutant by-products of traditional fossil-fuel based generation. Carbon emissions generated by SGs have been recognised as a large contributer to the worsening state of the climate, thus modern VRE and clean alternative sources of energy are being implemented as fast as possible. However, the technology lacks system voltage and frequency stability characteristics that has been historically provided by SGs. The inertia of huge rotating masses within SGs inherently act to mechanically resist change in the power system, reducing harmful frequency and voltage angle synchronisation dynamics between the generating plant and the grid during contingency events Shahab Karrari and Löckel (Aug. 2024). The reduction of total inertia in a network as SGs retire has led to reduced stability within electrical systems as inertia has been the fundamental first line of defence against instability, thus risking further contingency situations, such as under-frequency load shedding (UFLS), blackouts and potentially total system islanding (AEMO (2024)),as exemplified by the 2016 South Australian event. Inertia is the only factor limiting the rate of change of frequency (RoCoF) Shahab Karrari and Löckel (Aug. 2024))and thus heavily influences the magnitude of frequency nadir/zenith under contingency events, thus synthetic inertia has an important role to play in securing energy systems for the future. For the grid frequency to remain constant, total generation and consumption of active power must match, otherwise the grid frequency will deviate according to the swing equation (1). Inertia provided by SGs is dependant on the angular mismatch ($\frac{d\delta}{dt}$) between the grid and the SG, determined by the resultant change in coupling of electrical (P_e) and mechanical (P_m) power whereupon a contingency, the SG compensates for the power imbalance by releasing or absorbing energy. H represents the inertia constant of the generator - the amount of time in seconds the SG can provide its rated power from stored kinetic energy, a damping term proportional to the change in rotor angle for attenuating oscillation and w_0 stipulating the nominal grid frequency.

$$\frac{2H}{w_0}\frac{d^2\delta}{dt^2} = P_m - P_e - D\frac{d\delta}{dt} \tag{1}$$

Power converters have various control systems designed to imitate a near-instant power injection in the place of a SGs inherent inertial response.

1.1 Inverter-Based Resources

Inverter-Based resources (wind, solar, energy storage systems etc.) are interfaced to the grid through power electronics as opposed to an electro-mechanical coupling in SGs. Consequently, IBRs are rated for lower current carrying capability due to stricter current constraints - IBR's 1.2-2 p.u whereas SG's typically possess 6-8 p.u of over current capability (Jingyang Fang (2025)), however, as a consequence of their reduction there is an increasing demand for grid-forming (GFM) converter resources with ability for voltage and frequency stability in the grid (Shahab Karrari and Löckel (Aug. 2024)). Additionally, due to IBR's inherently lower current limits, the grid's short circuit ratio is declining with further IBR penetration. IBRs regulate the voltage amplitude and frequency by controlling active and reactive power delivered to the grid at the point of connection (POC). Grid-connected IBRs generally fall into two broad categories; Grid-Following (GFL) - in which the converter operates as a current source, and Grid-Forming (GFM) - operating as an independent voltage source with accompanying voltage phasor internally maintained by a variable current. However, a grey area overlaps these control modes, classified as grid-supporting by Joan Roacbert and Rodriguez (2012). Unlike a GFM inverter, which acts as an independent voltage source, these converters use different control architectures, such as droop control, to provide a rapid, inertia-like fast frequency response (FFR). Currently, GFL converters dominate IBR generation in power systems, utilising a Phase-Locked Loop (PLL) to synchronise with the grid's voltage waveform using and thus provide active or reactive power support. The measurement dependency of PLLs is precisely why GFL converters cannot provide an independent, instantaneous Synthetic Inertial Response (SIR) under contingency; the grid voltage becomes unstable, the PLL struggles to track the distorted reference thus forcing the inverter to reduce output or trip offline and thereby exacerbating the system instability. GFM converters act as a voltage source behind an impedance, capable of producing a SIR at the POC when grid synchronised. Instead of following the grid reference phasor, a GFM inverter independently forms its own internal voltage phasor and dynamically varies its current output to maintain it, a capability achieved through advanced control strategies like the Virtual Synchronous Machine (VSM). This approach allows a GFM-coupled BESS to imitate the inherent power injection of an SG with an extremely fast response time. In Australia, a fleet of pioneering GFM BESS projects funded by the Australian Renewable Energy Agency (ARENA) has successfully demonstrated the capability of large-scale GFM BESS to provide these critical grid services. Projects such as the Hornsdale Power Reserve Expansion (HPRX) and the Wallgrove Grid Battery (WGB) have been instrumental in proving the technology's effectiveness in inertial services and have driven regulatory change, such as the removal of restrictive grid access standards that previously hindered GFM deployment (ARENA (Jun 2025)).

1.2 Battery Energy Storage System (BESS)

BESS are an IBR that allows storage and discharge of energy through internal electrochemical reactions. Many types of battery chemistries are available with lithium-ion commonly deployed in BESS due to high power density and competitive cost. BESS scaled for grid services are capable of a range of applications and is specifically suited to FCAS due to their fast response time and high energy capacity, making them a suitable technology to provide synthetic inertia. BESS are specified by their rated power capacity (MW), energy storage capacity (MWh) and overload capacity. The overload capability is the BESS's ability to briefly exceed its nominal current rating, critical to enabling the rapid injection of active/reactive power required for an effective inertial response. Additionally, charging and discharging performance of BESS SIR is dependant on instantaneous battery state of charge (SoC), stipulating the volume of SIR available as the SoC denotes the available energy the BESS can contribute to the grid. Therefore, for the BESS to provide a SIR, suitable power buffer is required and can be achieved with one or more methods; maintaining power headroom by restricting the operating region of the GFM BESS, using an inverter with high overload capacity or over-sizing the GFM BESS for the expected contingency size. The total energy volume of the SIR a BESS can provide is fundamentally dependent on its SoC, which dictates the available energy for either charging or discharging. The extremely fast response offered by a GFM BESS is constrained by hardware operational limits like SoC and inverter current ratings, thus this report seeks to identify what characteristics allow GFM BESS to provide a SIR and the interplay of these parameters influencing a negative or positive impact on the power system.

2 Key factors contributing to synthetic inertia

A comprehensive literature review was conducted to assess the state of the field of grid-scale power converter technology, consisting of research articles, government documents and industry white papers to investigate BESS technology operational limits and characteristics that enable synthetic inertia provision. All BESS are connected to the grid through power electronics converter, utilising varying control algorithms depending on the services required in the system (Macit Tozak and Hayes (Jan 2024)). GFL inverter's control algorithms are reliant on phase-locked loop (PLL) to maintain a synchronised current to the grid's voltage phase (Joan Roacbert and Rodriguez (2012)), resulting in the converter not resisting but following, which can contribute to exacerbating disturbances. This PLL grid voltage measurement can be impacted by factors such as the impedance of the transmission line and noisy readings seen by the converter controls, distorting the BESS response. However, GFM converters have an advantage in providing a SIR as they do not require an external measurement to catalyse response to a contingency event, owing to their voltage source behaviour'. The inverter instantaneous active power is compared to the active power set-point and the converter acts accordingly to bridge the difference. For GFM converters, numerous control algorithms have been developed to provide voltage and frequency stability, with examples such as virtual synchronous machine (VSM) providing inertial response and frequency support, along with multi-loop droop control which can provide frequency support proportional to the frequency deviation, arresting the frequency nadir (Macit Tozak and Hayes (Jan 2024)). While both are GFM strategies, their control targets are fundamentally different: VSM provides an inertial response based on RoCoF, whereas droop control provides primary frequency support based on frequency deviation (Wei Du and Kundu (Jun 2020)) to provide an inertia-like response. Rick Wallace Kenyon and Hodge (2021) verified a PSCAD model consisting of GFM and GFL converters in a modified zero-inertia 9 bus IEEE test system, showcasing its ability to regulate frequency and power imbalance and settle 6 seconds faster than SGs in the same scenario. VSM control emulates the swing equation allowing the converter to provide an active power response proportional to the RoCoF in equation 2. This response can be fine-tuned by altering the inertia constant H and damping coefficient D.

$$\frac{2H}{f_0}\frac{df}{dt} = \Delta P \tag{2}$$

Renewable energy sources and power electronics have been prevalent in microgrids and as distributed energy sources since their introduction, with examples such as the Flinders Island VRE microgrid (Kutaiba S. El-Bidairi (2020)), however, it has not been until the last decade that power converters and energy storage devices have begun to be designed for grid-scale network stability regulation. There is an increasing demand for GFM converter capability for voltage and frequency stability (Shahab Karrari and Löckel (Aug. 2024)), leading to a recent rapid deployment and planning of GFM BESS projects in Australia - Hornsdale Power Reserve (HPRX), upgraded in 2019 and Wallgrove Battery (WGB) for additional synthetic inertia. It has been demonstrated that operational GFM BESS are capable of providing a SIR within the inertial window defined by AEMO ($500ms$) (AEMO (May 2023)), where key frequency response metrics include: RoCoF, frequency nadir, active power response volume, and settling time. It is imperative a BESS SIR acts near immediately and provides a stable response to worse case contingencies. For example, stable frequency operation is defined in NGESOs grid code GC1037 as a RoCoF of $1Hz/s$. Similar grid-scale projects are also being pioneered in the UK as a part of the national Stability Pathfinder program (M. Hishida and Knobloch (2023)), for example the Kilmarnock GFM BESS, a $300MW$ (rated $200MW$) BESS designed to provide $1500MWs$ of inertia to the UK system, including a maximum overloading capacity of 0.3 p.u. It was found SIR is limited by the current carrying capacity and short-term overloading capacity of the BESS inverters (M. Hishida and Knobloch (2023)), indicating that when designing a BESS for a SIR these characteristics can define an appropriate BESS response.

2.1 BESS SIR effect on state of charge

SoC of the BESS is defined as the ratio of available energy within the BESS at a point to its nominal rated capacity (equation 3). E (Wh) is the BESS rated energy and h converts Wh to Joules (Kutaiba S. El-Bidairi (2020)).

$$SoC_{BESS} = \frac{E_{available}}{E_{rated}} \tag{3}$$

The SoC of a BESS changes following a contingency due to active power transfer between the BESS and grid. A high RoCoF event demands a substantial and rapid power injection which can lead to a significant effect on SoC dynamics (Alhejaj and Gonzalez-Longatt (Sep. 2016)). To ensure the BESS can provide a sustained and bidirectional inertial response—injecting power during under-frequency events or absorbing power during over-frequency events— there exists an optimal SOC the BESS should operate at to allow headroom for a bi-directional SIR. If the available SoC is insufficient to support a high-volume SIR, the BESS cannot provide a sufficient active power response, exacerbating instability in the system and increase the risk of cascading contingencies throughout the network. Consequently, a reliable inertial response requires a minimum SoC equal to their rating for SIR.

While SoC is a critical constraint for sustained energy delivery, its impact on the brief inertial response, which occurs within the first few hundred milliseconds, can be considered negligible. A more immediate and critical constraint is the BESS's pre-disturbance operating point(its level of charging or discharging power).

2.2 BESS SIR and effect on angle stability

Rotor angle stability in IBRs denotes the mismatch in voltage angle at the inverter terminals and the grid POC (Macit Tozak and Hayes (Jan 2024)). In the GFM BESS case, high volumes of power exchange between the inverter and the grid can create oscillations of the frequency outside determined operational frequency limits (Kundar (Aug. 2004)). Synchronisation of a BESS to the grid through GFM control is achieved by applying the reduced swing equation where damping terms are neglected (M. Hishida and Knobloch (2023)), to determine the internal reference angle (θ) based on a power mismatch between P^* (power setpoint) and P (measured Power). The active power output in relation to the angle difference (θ) between inverter and the grid is as follows (equation 4)(M. Hishida and Knobloch (2023));

$$\ddot{\theta} = \frac{P^* - P}{2H} \tag{4}$$

The angle mismatch before and after a contingency event differs; however, consequent oscillations in the angle should not increase after a BESS SIR, as GFM converters alter their current and power output in seeking a new stable operating point for the system. Thus, a BESS SIR should have an acceptable effect on the angle between the grid and converter so that further instabilities in the system are not exacerbated (AEMO (May 2023)). It has been found through recent ARENA experience that GFM converters show a reduced angle sensitivity to changes in power flows (Badrzadeh et al. (2024)). There is no recognised industry consensus of a maximum volume of inertia response that does not induce angle instability, as historically many OEMs have designed GFM converters for other frequency services.

2.3 BESS overloading capability

Inverter overloading capability denotes the maximum apparent power (in p.u.) the inverter can withstand under high load conditions in response to variable network contingency sizes. A higher inverter maximum current capability allows the BESS to inject and absorb active power higher than BESS rated capacity, with current limited by the thermal capability of semi-conductor switches. A higher control inertia constant, H, requires a higher designed overload capability of GFM inverter hardware as a BESS SIR is proportional to H as per equation (1). Overloading capability is designed to withstand severe frequency disturbances considered in the network. For inertia services, GFM converters have a maximum H for the rated capability of the BESS, however as SIR is provided in addition to other services, GFM BESS must be designed in conjunction with overload capability to be able to provide SIR during maximum import/export conditions (AEMO (May 2023)). NGESO defines inertia for H in GFM converters as per (5), dependant on BESS rating, S_{rating}, ΔP ,designed active power change during a frequency event, and f_o denoting nominal network frequency (M. Hishida and Knobloch (2023)):

$$H = \frac{\Delta P * f_o}{2 * S_{rating} * RoCoF} \tag{5}$$

Equation (5) can be rearranged for ΔP to determine the active power volume required for SIR based upon a set H and the RoCoF operational limits. In the Kilmarnock BESS operation, ΔP is equal to S_{rating} and thus $H = 25s$ (M. Hishida and Knobloch (2023)). To determine the overload capability whilst keeping within specified RoCoF operating limits, the largest disturbance should be considered and H can be tuned accordingly (typically in the range of 130) to provide a SIR. The GFMI's internal voltage phasor is maintained during disturbances as the current phasor changes instantaneously, and an associated rapid increase in converter current could jeopardise the hardware components due to overheating (Macit Tozak and Hayes (Jan 2024)). Overloading capability is primarily dictated by the thermal management of the semiconductor switches. However, a more fundamental constraint lies in the control system's stability boundary: an excessively high H-value can lead to instability, such as oscillations or runaway if the system lacks sufficient damping (Alhejaj and Gonzalez-Longatt (Sep. 2016)). Thus, the BESS SIR should have an acceptable effect on system stability, and a stable maximum volume of inertia response must be identified.

2.4 Delay in synthetic inertial response

RoCoF detected at the PoC between the network and GFMI is one method to determine the switching response time of a BESS input volume for a SIR. The intention is to provide as inherent a response as possible to replicate a synchronous inertial response. For a synchronous machine, this initial response is an instantaneous, physical reaction with near-zero delay; however, the synthesized response of a GFM BESS is subject to inherent delays.Identified contributions to delaying a SIR of the inverters include the delay in measurement/system recognition of a disturbance that requires a SIR, including the switching command to GFMI delay (P. V. Brogan and Kubik (Jan. 2019)). The primary purpose of this inertial response is not to fully restore the frequency, but rather to arrest the rate of change of frequency (RoCoF), thereby buying crucial time for the subsequent, slower but more sustained frequency response (FFR) services to activate and stabilize the frequency. A timing window of $100ms$ (AEMO (May 2023)) is intended to isolate the inertial contribution of the GFM BESS, however there are fast frequency response (FFR) services that can act in opposition to a SIR to a disturbance within $200ms$ (AEMO (2024)), with the cross over of frequency services subject to future market designs and research. Furthermore, the 'quality' of the response within this window is critically shaped by the virtual damping parameter. A lower damping setting may allow for a more aggressive initial power ramp but risks causing overshoot and oscillations, while a higher damping setting can ensure a smoother, more stable response but may temper the initial reaction speed.

2.5 Control Strategies

The GFM capability described above is achieved through specific control strategies that emulate SG dynamics. The two most prominent methods are the Virtual Synchronous Machine (VSM) and multi-loop droop control. The Virtual Synchronous Machine (VSM) is the primary strategy for delivering SIR. It directly implements the swing equation within the inverter's control logic. By mathematically replicating the relationship between power imbalance, inertia, and frequency deviation, VSM enables the BESS to provide a truly instantaneous inertial response. However, this emulation introduces its own stability challenges. Specifically, dynamic interactions related to a high virtual inertia constant (H) or specific current limitation strategies (eg. Q/P priority) can create instability, particularly in weak electrical grids characterized by a low Short Circuit Ratio (SCR). This paradox, where the control parameter intended to enhance stability can itself become a source of oscillation, is a central investigation of this report. Multi-loop droop control, on the other hand, is fundamentally a mechanism for Primary Frequency Response (PFR), not instantaneous inertial response.This is the core mechanism used by the 'grid-supporting' converters mentioned previously. It enables a fast frequency response, but that is based on the magnitude of the frequency deviation, not its rate of change.It adjusts the BESS's power output in proportion to the magnitude of the frequency deviation from its nominal value (Australia is $50Hz$ and South Korea is 60Hz), rather than its rate of change. Because this frequency error must first exist and accumulate over time, the droop response is inherently slower, placing it outside the inertial time period. In advanced GFM architectures, the VSM provides the fast, inner-loop inertial response to absorb the initial shock, while the slower droop control acts as an outer loop to arrest the RoCoF. It is also possible to have dynamic control of parameters such as H and D for improved SIR stability during a contingency. However, this control is not yet defined by TSO's and implemented independently by original equipment manufacturers (OEM) (ARENA (Jun 2025))(Fauzan Hanif Jufri and Garniwa (Jun 2023))(Macit Tozak and Hayes (Jan 2024)).

3 Methodology

Time-domain simulations were performed to produce a quantitative study of the effect of different BESS control variables on the SIR produced. The simulations were performed on an open-source Matlab/Simulink GFM model. The model consists of a $1MVA$, $415V$, $50Hz$ three-phase GFM-VSM converter connected to a $415V/11kV$ step-up transformer. Additionally, a local load is connected to the low side of the transformer. The high side of the transformer is connected to a transmission line (modelled as a series RL impedance), which then connects to an infinite bus (ideal voltage source representing the external grid).
The following changes were made from the original model; The ideal DC voltage source was replaced with the generic battery model (Simscape/Electrical/Sources) set to $1100V$, the size of the converter was increased from $500kVA$ to $1MVA$. The MATLAB BESS model is simplified in design with a single GFM converter, whereas operational BESS projects consist of multiple converters in parallel to increase power output. All tests were performed under the assumption that multiple converters(when tuned) will behave identically to a single larger converter.
All data were collected from the following tests. Frequency ramp test which consists of the infinite bus maintaining a nominal $50Hz$ for the first $10s$ while the simulation environment stabilises, then increases to $52Hz$ at a constant RoCoF of $2Hz/s$, frequency stays constant for 5 seconds, and finally the frequency decreases by $-1Hz/s$ until $47Hz$ is reached. The purpose of this test is to highlight how independent variables affect the stability and volume of the SIR and reactive power.
The load change test consists of the same stablising period, but at $10s$ the local load changes from $0.4pu$ to $0.7pu$ of active power. The purpose of this test is to simulate a contingency event, drawing relationships between the independent variables and the PoC frequency RoCoF, nadir, and nadir time. Overloading Capability test is similar to the load change test, but instead of having a constant active power step change of 0.3 p.u the load step change was changed to find the point where the GFM-BESS became unstable.
The RoCoF range test is similar to the frequency ramp test, but instead of a constant frequency ramp, the RoCoF was changed to find the limits of stable operation.
The transmission line test varied the length of the transmission line that connects the infinite bus to the GFM-BESS. The angle jump test consists of the infinte bus voltage phase jumping by different values, the phase does not recover during the test.
The maximum angle jump test varies the H and D parameters of the inverter controls and notes down the largest angle jump (not recovering) the GFM-BESS can handle before becoming unstable.
RoCoF was estimated by considering only the frequency nadir point and disturbance point of the frequency-time graph. This estimation only captures the average RoCoF over this time period, the maximum RoCoF will most likely exceed this estimation.

$$Ro\bar{C}oF = \frac{f_{nadir} - f_0}{t_{nadir} - t_{disturbance}} \tag{6}$$

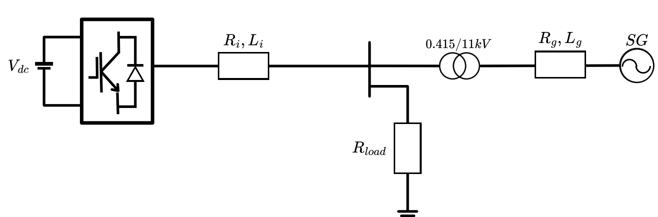


Figure 1: Case study system diagram used for simulation studies

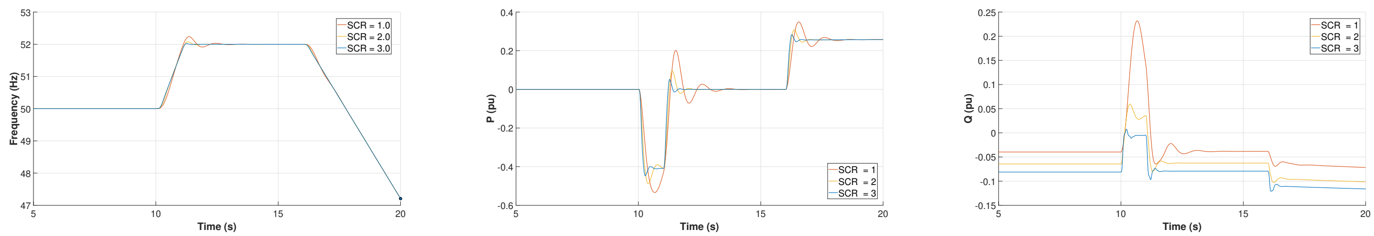
Symbol	Description	Value
H	Inertia Constant	2
D	Damping Coefficient	1
OLC	Overload Capabililty	0 p.u
SCR	Grid Short circuit ratio	2.5
XR	Grid X/R Ratio	4
V_{dc}	DC Voltage	1100 V
Q	Rated Capacity	1 MWh
f_g	Grid Frequency	50 Hz
l	Line Length	2 km
R_g	Line Resistance	1.0117 Ω/km
L_g	Line Reactance	0.025 Ω/km
R_i	Inverter Filter Resistance	0.1 Ω
L_i	Inverter Filter Reactance	0.1722 Ω

Table 1: Base Case Parameters

4 Results and Discussion

4.1 Grid SCR

The short-circuit ratio of the grid defines the strength of the grid (IEEE standard 1204-1197), with an SCR > 3 considered a strong grid, however GFM converters perform better in weak grids due to their self-synchronisation characteristics
Lower SCR is associated with weaker power systems, and so can be more easily influenced, this can be seen in Figure 2.a where the BESS can resist the frequency ramp more effectively in lower SCR systems. Observing Figure 2.b the SIR overshoots and oscillates more in lower SCR systems. SCR of 1 results in two to three oscillations before reaching steady state, whereas a SCR of 3 oscillates just once. Larger overshoots can also be observed as the SCR of 1 has an overshoot approximately 0.15 p.u greater than the SCR of 3 case. As a result, lower SCR power systems limit the range of RoCoF the BESS can respond to without the overshoot exceeding the inverter current limits.
Additionally, Figure 2.c highlights that higher SCR systems require less reactive power to maintain voltage stability, leaving a greater power reserve for SIR. Therefore, as SCR of the power system increases, so does the SIR effectiveness.

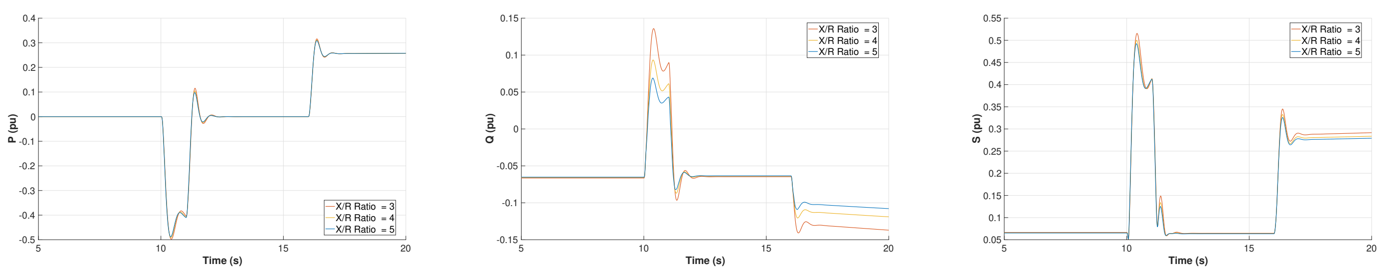


(a) Frequency Ramp - PoC Frequency (b) Frequency Ramp - Active Power (c) Frequency Ramp - Reactive Power

Figure 2: Affect of Grid SCR

4.2 X/R Ratio

The grid X/R ratio is defined as the ratio of reactance (X) to the resistance (R) of the connected system. Observing the results from the frequency ramp test, Figure 3.a and 3.b, the active power response is nearly identical across all X/R ratios, whereas the reactive power differs. This is significant as lower X/R ratio systems require a larger reactive power response to facilitate the same SIR. As a consequence, for the same GFM-BESS, lower X/R ratio systems will require a larger reserve of the inverter power capacity for reactive power, thus reducing the headroom available and limiting SIR.

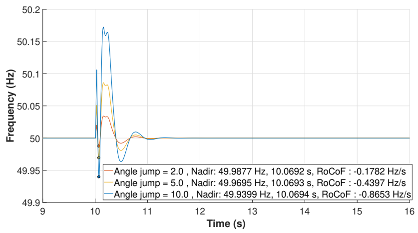


(a) Frequency Ramp - Active Power (b) Frequency Ramp - Reactive Power (c) Frequency Ramp - Apparent Power

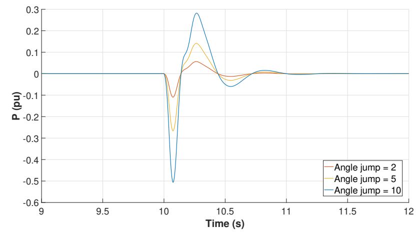
Figure 3: Affect of grid X/R Ratio

4.3 Angle Jump

A larger angle deviation indicates a larger disturbance to the systems equilibrium, directly influencing the amount of active power that is injected during an active power response confirmed in Figure 4.b where an angle jump test of 10 degrees shows the largest frequency deviation of the system and the longest settling time. The frequency response exhibited by the multiple angle jump tests suggest a proportional relationship between an increase in angle jump during a disturbance and the resultant frequency deviation under such conditions provided the system is stable. The active power response to an angle mismatch between the converter internal setpoint and the grid PoC follows a similar proportionality to the angle deviation as the frequency response. An angle jump of 10 degrees shows a magnitude 5 times that of an angle jump of 2 degrees as the system stabilises to its new angle setpoint. From the active power and frequency responses observed from an angle mismatch, the magnitude of angle deviation does not impact the settling time of the system to its new equilibrium point, affecting the magnitude of active power oscillations and frequency deviations experienced by the model. Thus further simulations were performed to assess the maximum angle jump the system could experience whilst maintaining stability with varying SIR magnitudes, dictated by combination of inertia constant and damping coefficient control settings. Increasing the inertia constant whilst maintaining a constant damping coefficient reduced the maximum angle change the system remained stable under likely due to the increased SIR associated with an increased H value. Conversely, maintaining a constant H whilst increasing the damping coefficient resulted in a lower magnitude but increase in system robustness and tolerance to an angle change, compared to H setpoint change, an expected response due to the role of damping in a power system.



(a) Load Change - PoC Frequency



(b) Load Change - Active Power

H, D	**Angle jump**
$1, 1$	13.9°
$2, 1$	13.75°
$4, 1$	13.3°
$2, 0.5$	13.7°
$2, 2$	13.8°

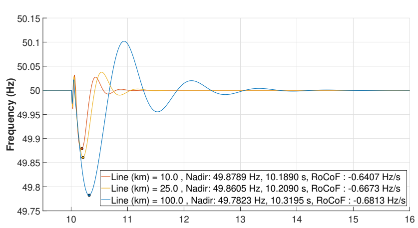
(c) Maximum angle jump (Resolution of 0.05)

Figure 4: Angle Jump

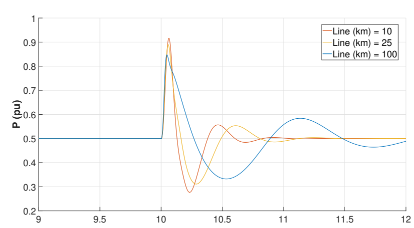
4.4 Transmission line length

If the BESS is further away from the converter terminals at the plant, there is an expected increase in losses to the volume of power seen at the grid POC for an SIR. The effect of increasing the length of the transmission line between the transformer and the grid POC was investigated in the load change scenario.

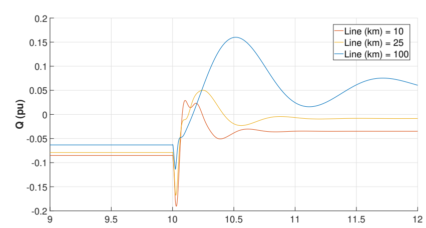
The load change test shows the relationship between an increasing transmission line length and the active power and frequency responses by the BESS system. After the disturbance occurs the BESS injects active power according to its settings in the form of SIR. Since the load only demanded 0.3 p.u. of active power all the remaining power gets pushed towards the grid with the inductive nature of the transmission lines causes oscillations and increased frequency settling time. As expected, with increasing transmission line length resulted higher active power losses and a slower overall frequency response were observed in the model, indicated by figures (5a) and (5b). An increase in RoCoF, frequency nadir and time taken to reach the frequency nadir with an increasing transmission line length was observed (figure 5a), with the system taking an additional 30ms to be arrested at the nadir time when the transmission line length is 100km. In constrast, transmission line length of 10km allows the frequency deviation to be arrested in a sub-200ms window, whilst also contributing the highest volume of power at the POC (figure 6b), verifying that the shorter the transmission line the lower the impedance along the length on the line, thus increasing the active power contribution of the SIR seen at the grid POC.



(a) Load Change - PoC Frequency



(b) Load Change - Active Power

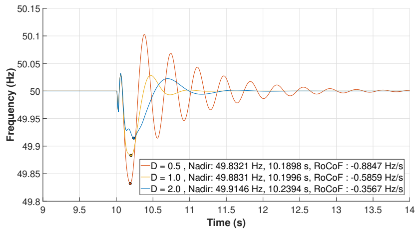


(c) Load Change - Reactive Power

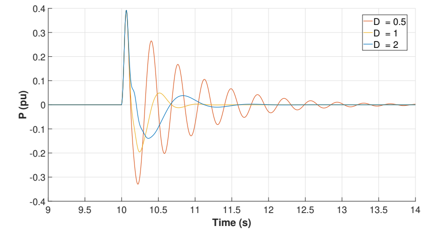
Figure 5: Affect of Transmission Line Length

4.5 Damping Coefficient (D)

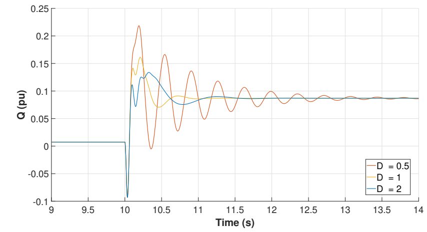
Damping is critical characteristic of power systems that restricts oscillations in the frequency resulting from a change in the active power setting. Figures 6a/b show the frequency and active power responses from the BESS after an increase in local load from 0.4p.u to 0.7p.u., attempting to simulate the SIR response of the BESS and analyse how the damping coefficient affects the RoCoF (eqn 6), frequency nadir and the frequency nadir time. The results support expected effect damping has on the system, as in figure 6a for D=2, the magnitude of the frequency nadir and the RoCoF are the lowest of the coefficients tested, indicating the effect damping has on a load change in the system. However, the frequency nadir was reached the slowest for D=2 due to initial oscillations as the system initially resisted the frequency change when the load changed, however settles the frequency much faster with fewer oscillations than D=1, 0.5. The active power responds with sufficiency within the expected inertial window, with the peak response approximately 100ms after the load change occurs. A maximum active power discharge of 0.39p.u. occurs for all tested D coefficients, matching the required SIR for the load change, however the stark difference observed between values is the highly damped response when D=2 compared to the highly oscillating response when D=0.5. Observing Figure 6.b as D increases, the SIR shows reduced oscillations and a lower overshoot. This is evident as the D value of 0.5 results in active power oscillations prolonging past the 14 second mark. Increasing the D to 1 and 2 reduces these oscillations, settling just after 11.5 seconds and 11 seconds respectively. Furthermore, increasing D decreases the depth of the frequency nadir, demonstrating its FFR nature to arrest the frequency decline. Figure 6.a shows that a D of 0.5 results in a frequency nadir of 49.8321 Hz, increasing the D to 1 and 2 decreases the frequency nadir depth by 30.4% and 26.9%, to 49.8831 Hz and 49.9146 Hz respectively.



(a) Load Change - PoC Frequency



(b) Load Change - Active Power

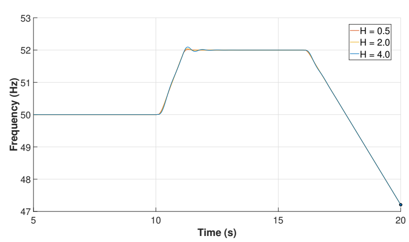


(c) Load Change - Reactive Power

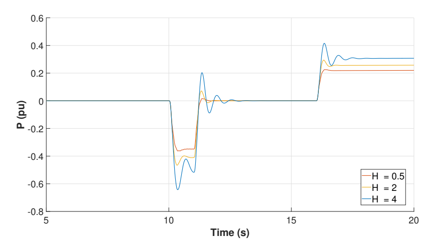
Figure 6: Affect of D

4.6 Inertial Constant (H)

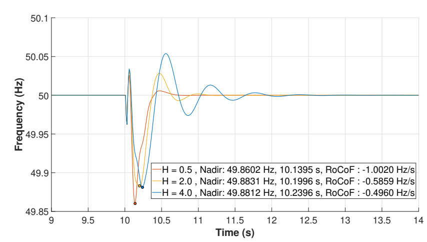
It is observed that the active power output of the GFM-BESS reacts opposite in sign, but proportional in magnitude to the RoCoF of the infinite bus. This verifies the expected behaviour of a SIR. From Figure 4.b it is clear that as H increases the SIR demonstrates larger overshoot and longer oscillations before reaching a steady-state value. For $H = 0.5$ the SIR experiences a brief overshoot then settles to its steady-state value without oscillating. For $H = 2, 4$ the overshoot is greater as H increases, and the oscillations go to one to two cycles and two to three cycles respectively. Furthermore, as H increases the steady-state SIR reached is larger. It is important to note that although the SIR increase with H it does so in a complex and non-linear relationship, evident as the SIR does not quadruple as H goes from 0.5 to 2, then double from 2 to 4.
The same characteristics can be observed in the load change test, where larger H values lead to larger SIR, but longer recovery time from oscillations. Immediately following the disturbance the SIR begins to ramp (Figure 7.b), taking approximately 60ms to reach its maximum output. This ramping time is consistent with all H values, indicating a fixed time delay. As the battery model is simplified to not simulate current ramping dynamics the likely causes are a combination of PoC current and voltage measurement delays and virtual rotor angle dynamics from swing equation calculations.
Figure 7.c shows the PoC frequency response during the load change test; as H increases from 0.5 to 2 the frequency nadir time extends by 0.0601 seconds, from H equals 2 to 4 the nadir time extends by a further 0.04 seconds. Furthermore, the frequency nadir itself does not vary significantly with H. From equation (reference above) it can be seen that as H increases the RoCoF at the PoC reduces.
Increasing H results in an increased volume of SIR, effectively reducing system RoCoF during contingency events, although, this comes at the cost of increased settling time and larger oscillations.



(a) Frequency Ramp - PoC Frequency



(b) Frequency Ramp - Active Power



(c) Load Change - PoC Frequency

Figure 7: Affect of H

4.7 Overloading capability (OLC)

An OLC value of 0 p.u represents a BESS that cannot exceed its nominal rated current. The BESS initially discharges 0.5 p.u of active power into a system load of 0.4 p.u. A contingency is applied at $t = 10$ s. Without additional OLC, the BESS remains stable up to a contingency size of 0.75 p.u.
Increasing the OLC enhances the current-carrying capability of the BESS, enabling a higher active power response. For instance, an OLC of 0.1 p.u allows the BESS to withstand a contingency 0.25 p.u larger than in the base case. However, further increasing the OLC from 0.1 p.u to 0.3 p.u yields a smaller improvement, only an additional 0.1 p.u in contingency size.
This non-linear trend can be attributed to the reactive power demand associated with increased active power output. Although the BESS reactive power setpoint was zero and the load was purely resistive, reactive power exchange occurs between the BESS and the grid to maintain the PoC voltage. The larger the SIR, the more the voltage sags, and the greater the reactive power

required to sustain voltage stability. Consequently, when designing OLC for SIR, sufficient reactive power reserve must be considered to ensure stable voltage support during high active power response.

Overloading Capability (p.u)	Largest Contingency (p.u)
0	0.75
0.05	0.85
0.10	1.00
0.15	1.05
0.20	1.10
0.25	1.10
0.30	1.10

Table 2: Overloading Capability vs. Largest Contingency (Resolution of 0.05 p.u)

4.8 BESS Operating point (OP)

This operating point directly dictates the instantaneous headroom available for the BESS to provide power upwards or downwards, thus limiting the range of RoCoF it can respond to. Figure 8 clearly illustrates this relationship:

1. Nearly-Symmetrical Response (OP = 0): When the BESS is free (at 0 p.u OP, green line), its response capability is at its most symmetrical state. It can respond effectively to both a frequency decline (requiring discharge) and a frequency rise (requiring charge), with a responsive RoCoF range of approximately -4.5 Hz/s to $+4.0$ Hz/s. The minor asymmetry observed is typically a result of the system's inherent efficiency differences between discharging and charging. Discharging is often slightly more efficient than charging due to unavoidable power conversion losses in the BESS inverter,transformers.
2. Asymmetrical Response while discharging (OP $>$ 0): When the BESS is already discharging pre-disturbance (at 0.5 p.u, blue line), its response capability becomes significantly asymmetrical. It has substantial headroom to respond upwards (charge), tolerating a positive RoCoF up to $+4.85$ Hz/s, but its headroom to respond downwards (discharge further) is compressed, only allowing it to handle a negative RoCoF of -2.3 Hz/s. This is because a portion of its discharge capability is already in use.
3. Asymmetrical Response while Charging (OP $<$ 0): when the BESS is charging pre-disturbance (at -0.5 p.u, red line), the situation is reversed. It has substantial headroom to respond downwards (discharge), tolerating a negative RoCoF up to -4.8 Hz/s, but its headroom to respond upwards (charge further) is severely compressed, allowing it to handle only a positive RoCoF of $+1.8$ Hz/s.

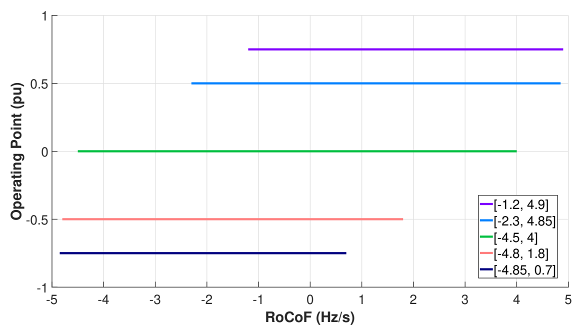


Figure 8: RoCoF range GFM-BESS can respond to varying Operating Point

5 Conclusion

The performance of a Grid-Forming Battery Energy Storage System (GFM-BESS) providing a Synthetic Inertial Response (SIR) is governed by a complex interplay between control parameters, hardware limitations, and grid conditions rather than by any single factor. BESS units equipped with Virtual Synchronous Machine (VSM) control can emulate synchronous generator behavior, where the inertia constant H and damping coefficient D parameters shape the magnitude, overshoot, and settling time of the SIR. The inverter's current-carrying capability, overloading capacity, and operating point determine the instantaneous power headroom available to deliver the response. Grid characteristics, particularly the short-circuit ratio (SCR) and X/R ratio—further, influence the voltage stability and reactive power requirements, thereby constraining the active power available for inertial support. Additionally, sufficient state of charge (SoC) headroom is required to ensure adequate energy reserves for either charging or discharging during SIR. Consequently, the inertial performance of a GFM-BESS cannot be regarded as fixed, its effectiveness is dynamically shaped by its controls tuning, BESS operational state, and the electrical strength of the grid to which it is connected.

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